

Final Project - Applied Statistics for Engineering and Scientists II

Group 6

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1 Part I - Concrete Compressive Strength

1.1 Exploratory Data Analysis

Variable Description

The Concrete Compressive Strength dataset (Yeh 1998) describes concrete mixtures based on their material composition and age. The objective is to explain and predict concrete compressive strength, which represents the amount of compressive load the concrete can withstand before failure.

- **Cement:** Amount of cement in the concrete mixture (kg/m³).
- **BlastFurnaceSlag:** Amount of blast-furnace slag in the mixture (kg/m³).
- **FlyAsh:** Amount of fly ash in the mixture (kg/m³).
- **Water:** Amount of water used in the mixture (kg/m³).
- **Superplasticizer:** Amount of superplasticizer used in the mixture (kg/m³).
- **CoarseAggregate:** Amount of coarse aggregate, such as gravel or crushed stone (kg/m³).
- **FineAggregate:** Amount of fine aggregate, typically sand (kg/m³).
- **Age:** Age of the concrete at the time of strength measurement, measured in days.
- **Strength:** Concrete compressive strength measured in MPa. This is the response variable to be predicted.

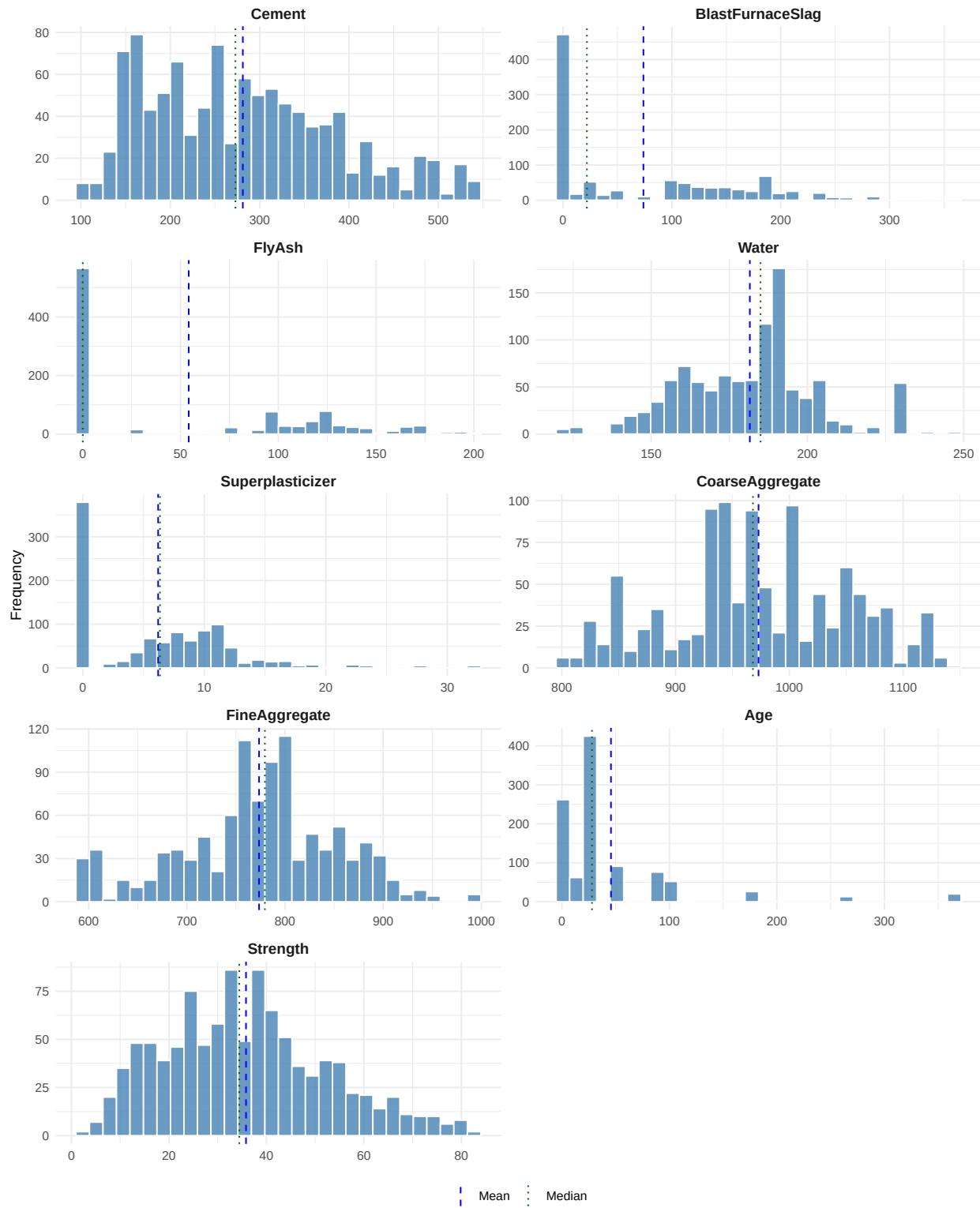
Dimension: 1030 x 9 | Total missing values: 0

The dataset contains 1,030 observations and 9 numerical variables, consisting of eight predictors and one continuous response variable, concrete compressive strength. No missing values were detected in any variable, so no missing-value imputation is required.

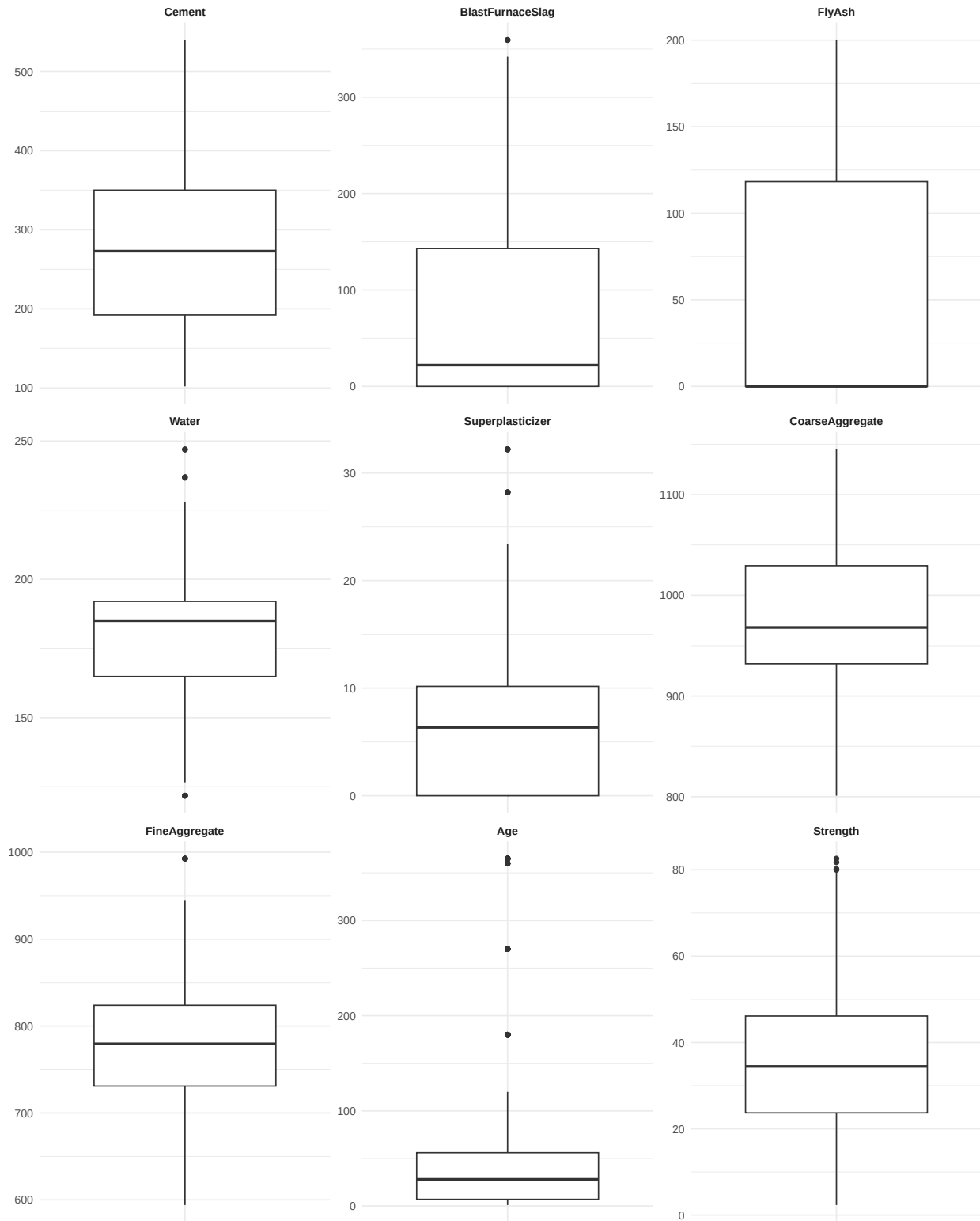
Distribution and Potential Outliers

- **Age** is strongly right-skewed and contains several unusually large observations, suggesting that a transformation may be worth investigating.
- **BlastFurnaceSlag** and **Superplasticizer** are strongly right-skewed with many zero values. These zeros likely represent mixtures where the ingredient was not used rather than missing data.
- **FlyAsh** also has a large concentration at zero, with a right-skewed distribution among the positive observations.
- **Water** contains a few potential low and high outliers, while **FineAggregate** has one potential high outlier.
- **Strength** is moderately right-skewed with a small number of unusually high strength observations.
- **Cement** and **CoarseAggregate** have relatively balanced distributions with no obvious extreme outliers.
- Potential outliers identified from the boxplots are not automatically removed; their validity will be investigated during the data-cleaning stage.

Distribution of Variables



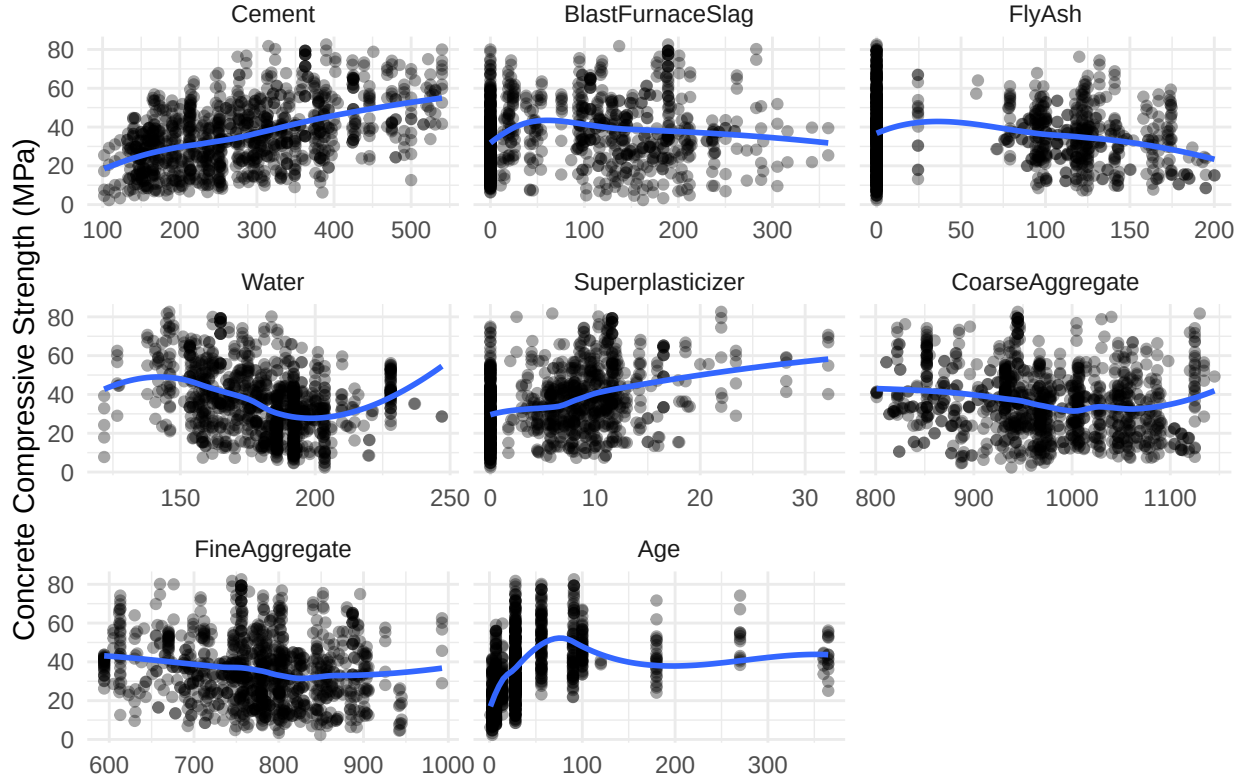
Boxplots of Variables



Pairwise Relationships

- **Cement** shows a clear positive and approximately linear relationship with Strength.

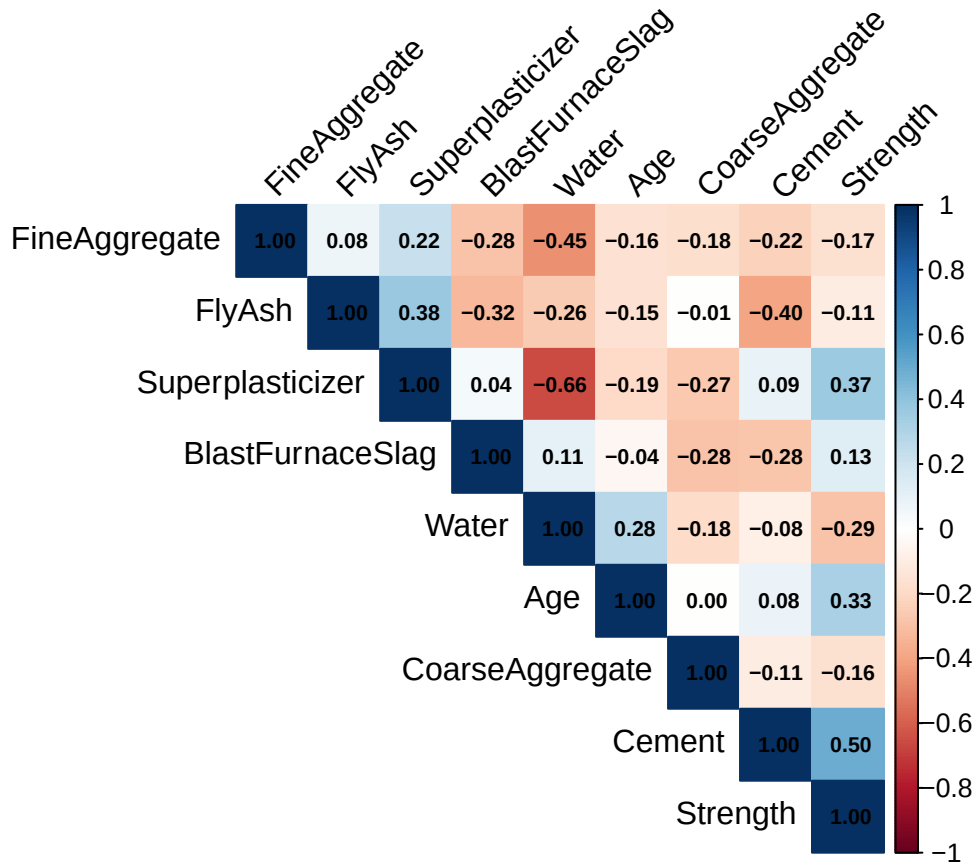
Relationships Between Predictors and Concrete Strength



- **Age** exhibits pronounced nonlinearity, with Strength increasing rapidly at early ages and then levelling off; therefore, a log transformation of Age will be investigated.
- **Superplasticizer** shows a positive but mildly nonlinear association, while **Water** displays substantial curvature.
- **BlastFurnaceSlag**, **FlyAsh**, **CoarseAggregate**, and **FineAggregate** show weaker marginal relationships with Strength.
- Overall, the plots suggest that a purely linear specification may not fully capture all predictor-response relationships. These marginal plots are used to motivate further transformation checks.

Correlation Analysis

- **Cement** has the strongest positive linear correlation with Strength ($r = 0.50$).
- **Superplasticizer** ($r = 0.37$) and **Age** ($r = 0.33$) also show positive linear associations with Strength.
- **Water** has a negative correlation with Strength ($r = -0.29$), indicating that higher water content tends to be associated with lower compressive strength.
- **FlyAsh**, **CoarseAggregate**, **FineAggregate**, and **BlastFurnaceSlag** show relatively weak linear correlations with Strength.



- Among the predictors, the strongest pairwise correlation is between **Water** and **Superplasticizer** ($r = -0.66$), suggesting some potential collinearity.
- No predictor pair has an absolute correlation greater than 0.70, so there is no evidence of severe pairwise multicollinearity at this stage.

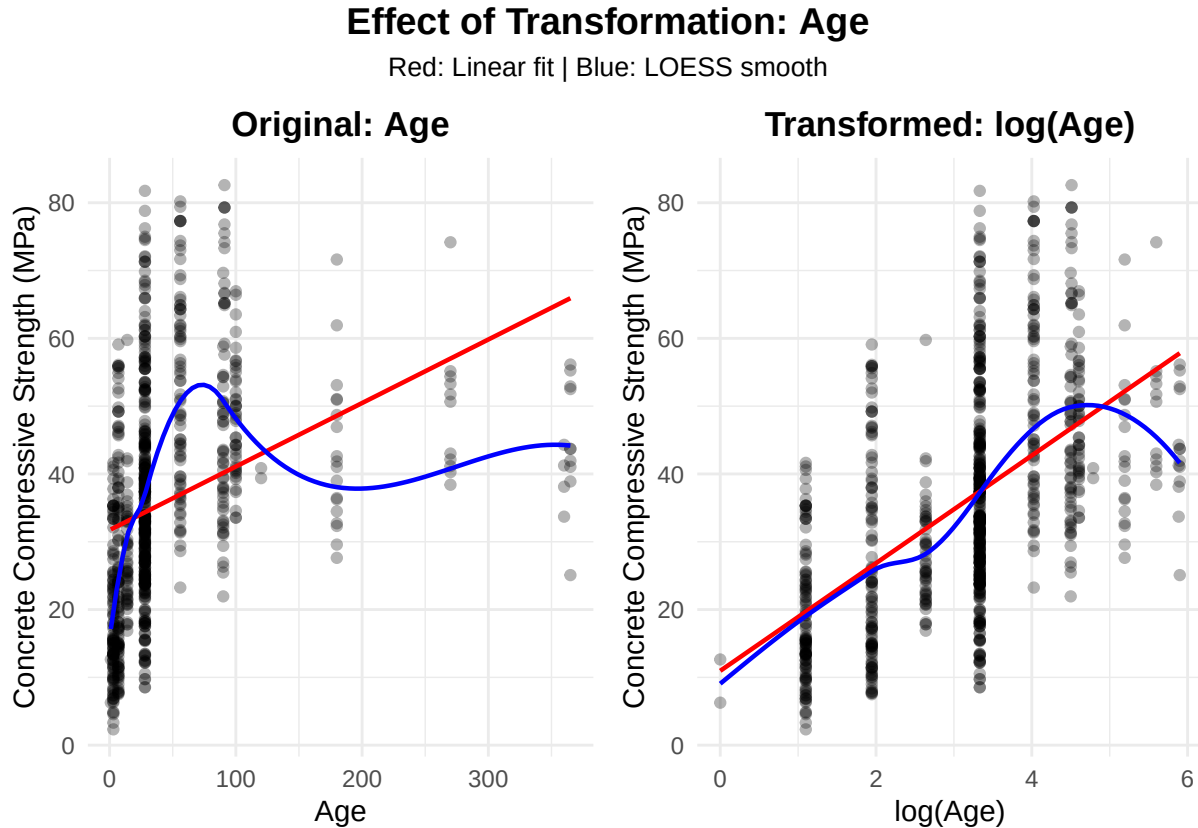
1.2 Data Cleaning

##	Variable	Lower_Bound	Upper_Bound	Number_of_Outliers	Percentage
## 1	Cement	-52.54	605.92	0	0.00
## 2	BlastFurnaceSlag	-213.75	356.25	1	0.12
## 3	FlyAsh	-177.28	295.47	0	0.00
## 4	Water	124.25	232.65	7	0.85
## 5	Superplasticizer	-15.24	25.40	5	0.61
## 6	CoarseAggregate	785.90	1175.50	0	0.00
## 7	FineAggregate	583.75	968.15	2	0.24
## 8	Age	-66.50	129.50	43	5.22
## 9	Strength	-10.08	79.98	3	0.36

- Potential outliers were identified from the training set using the $1.5 \times IQR$ rule.
- Most variables contained very few flagged observations. **Water**, **Superplasticizer**, **FineAggregate**, **BlastFurnaceSlag**, and **Strength** each had less than 1% of observations outside the IQR bounds.

- **Age** had the largest number of flagged observations (43 observations, 5.22%). However, its strongly right-skewed distribution suggests that these points mainly represent the upper tail of the age distribution rather than obvious data errors.
- The flagged observations were retained because no clear evidence indicated measurement or data-entry errors. Removing valid extreme concrete mixtures could discard useful information and bias the subsequent models.
- The skewness of **Age** will instead be addressed by investigating a log transformation.

Variable Transformation



Correlation with Strength:

Age: $r = 0.3383$; $\log(\text{Age})$: $r = 0.5585$

- **Age** was strongly right-skewed and showed a pronounced nonlinear relationship with Strength. After applying a logarithmic transformation, the relationship became substantially more linear and the Pearson correlation increased from $r = 0.338$ to $r = 0.558$. Therefore, $\log(\text{Age})$ was used in subsequent modelling.
- **Cement**, **Water**, **FlyAsh**, **BlastFurnaceSlag**, **Superplasticizer**, **CoarseAggregate**, and **FineAggregate** were otherwise retained on their original scales because there was insufficient evidence that transformation would improve the modelling relationship.
- **Strength** was retained on its original MPa scale to preserve direct interpretation of predictions and evaluation metrics.

- All original predictors are numeric, so no categorical encoding was required.

Feature Engineering

- Zero values in **BlastFurnaceSlag**, **FlyAsh**, and **Superplasticizer** indicate ingredient absence, so binary presence indicators were added while retaining the original dosage variables, allowing the models to distinguish between ingredient presence and dosage.
- A quadratic term, $Water^2$, was added because of the suggested curvature in the relationship between Water and concrete strength.

1.3 Feature Selection

Variance Inflation Factor (VIF) screening and stepwise selection were used as selection methods in this section for the following reasons:

- Most pairwise correlations among predictors are not severe ($|r| < 0.45$, except $r(\text{Water}, \text{Superplasticizer}) = -0.66$). However, pairwise correlation alone may not capture multicollinearity involving multiple predictors. Therefore, VIF is needed to assess the multicollinearity level of the predictors.
- VIF only assesses multicollinearity and does not determine which predictors should be retained in the final model. Therefore, stepwise selection is also used to identify a more parsimonious set of predictors based on model fit and complexity.

Table 1: Variance Inflation Factor (VIF) of Candidate Predictors

	Predictor	VIF	Assessment
4	Water	187.87	Very High
9	Water2	186.70	Very High
3	FlyAsh	15.94	Very High
2	BlastFurnaceSlag	10.57	Very High
11	FlyAshPresent	10.55	Very High
1	Cement	8.05	High
7	FineAggregate	7.58	High
6	CoarseAggregate	5.57	High
12	SuperplasticizerPresent	5.13	High
5	Superplasticizer	5.08	High
10	SlagPresent	3.41	Acceptable
8	LogAge	1.08	Acceptable

- The VIF results indicate that **LogAge** (1.08) and **SlagPresent** (3.41) have acceptable levels of multicollinearity, while several other predictors show high VIF values. In particular, **Water** (187.87) and **Water2** (186.70) have extremely high VIFs because **Water2** is directly derived from **Water**, creating strong dependence between the polynomial terms, which is expected.
- High VIFs are also observed for **FlyAsh**, **BlastFurnaceSlag**, and their presence indicators. This is expected in the results because the continuous variables and binary indicators capture related but distinct information: ingredient dosage and ingredient presence/absence, respectively.

Table 2: Feature Selection Results from Stepwise AIC

Predictor	Selected
Cement	Yes
BlastFurnaceSlag	Yes
FlyAsh	Yes
Water	Yes
Superplasticizer	Yes
CoarseAggregate	Yes
FineAggregate	Yes
LogAge	Yes
Water2	Yes
SlagPresent	No
FlyAshPresent	Yes
SuperplasticizerPresent	Yes

Stepwise selection

- Despite the high VIF values observed for several predictors, stepwise AIC retained most of them because removing these variables would reduce model complexity but lead to a poorer model fit.
- In contrast, **SlagPresent** was excluded despite having an acceptable VIF (3.41), indicating that its exclusion was not driven by multicollinearity but by its limited contribution to the overall model fit. This may also be explained by the presence of **BlastFurnaceSlag**, which already captures the dosage information of the same ingredient.
- The remaining presence indicators, **FlyAshPresent** and **SuperplasticizerPresent**, were retained, suggesting that they provided additional information beyond their corresponding continuous dosage variables.
- Therefore, **SlagPresent** was removed from the model, while the other selected predictors were retained.

1.4 Modeling with Regularization

The Linear Regression was considered as the baseline model and apply Ridge, Lasso, and Elastic Net as regularized alternatives in this section for the following purposes:

- Linear Regression: Used as the baseline model to provide a reference for evaluating the regularized models.
- Ridge Regression: Used to reduce the impact of multicollinearity by shrinking correlated predictors' coefficients without removing variables.
- Lasso Regression: Used for automatic variable selection by shrinking some coefficients exactly to zero.
- Elastic Net: Used to combine Ridge's stability with Lasso's variable selection, especially for correlated predictors.

1.4.1 Linear Regression

Table 3: OLS Baseline Model Coefficients

Predictor	Coefficient	P_value
(Intercept)	-20.8899	0.3454
Cement	0.1275	< 0.001
BlastFurnaceSlag	0.1076	< 0.001
FlyAsh	0.0340	0.0224
Water	-0.5862	< 0.001
Superplasticizer	-0.3184	< 0.001
CoarseAggregate	0.0246	< 0.001
FineAggregate	0.0294	< 0.001
LogAge	8.6173	< 0.001
Water2	0.0012	0.0034
FlyAshPresent	3.5841	0.0186
SuperplasticizerPresent	7.6162	< 0.001

- All predictors have $p < 0.05$, indicating statistically significant associations with Strength conditional on the other predictors. The relatively high VIF values, particularly for **Water** and **Water2**, should be considered when interpreting individual coefficients because multicollinearity can make coefficient estimates less stable.

1.4.2 Ridge regression

Table 4: Ridge Regression Coefficients

Predictor	Coefficient
(Intercept)	33.5491
Cement	0.0867
BlastFurnaceSlag	0.0592
FlyAsh	-0.0134
Water	-0.1475
Superplasticizer	-0.1066
CoarseAggregate	-0.0087
FineAggregate	-0.0146
LogAge	8.0877
Water2	-0.0003
FlyAshPresent	2.5448
SuperplasticizerPresent	6.8236

Compared with OLS, Ridge substantially shrinks the coefficients of several predictors, especially the highly collinear variables **Water**, **Water2** and **Superplasticizer**. Other continuous predictors, including **CoarseAggregate**, **FineAggregate**, **BlastFurnaceSlag** and **FlyAsh**, are also strongly reduced. Meanwhile, larger coefficients are retained for **LogAge**, **FlyAshPresent**, and **SuperplasticizerPresent**. This indicates a redistribution of predictive contribution among correlated variables, which helps Ridge reduce the impact of multicollinearity.

1.4.3 Lasso regression

Table 5: Lasso Regression Coefficients

Predictor	Coefficient
(Intercept)	-25.0561
Cement	0.1263
BlastFurnaceSlag	0.1061
FlyAsh	0.0322
Water	-0.5024
Superplasticizer	-0.3024
CoarseAggregate	0.0233
FineAggregate	0.0276
LogAge	8.6299
Water2	0.0009
FlyAshPresent	3.6165
SuperplasticizerPresent	7.4926

Compared with OLS, Lasso shrinks the coefficients of most predictors toward zero, with the strongest shrinkage observed for **Water**, **Water2**, **FlyAsh** and the aggregate variables. However, no coefficient is reduced to zero, so all predictors are retained in the model. Compared with Ridge, Lasso applies slightly less shrinkage overall, while preserving the same signs of the coefficients as OLS.

1.4.4 Elastic Net regression

Table 6: Best Elastic Net model

alpha	lambda	cv_mse
0.2	0.0051824	47.64014

Table 7: Elastic Net Regression Coefficients

Predictor	Coefficient
(Intercept)	-26.7288
Cement	0.1266
BlastFurnaceSlag	0.1064
FlyAsh	0.0327
Water	-0.4921
Superplasticizer	-0.3005
CoarseAggregate	0.0236
FineAggregate	0.0279
LogAge	8.6321
Water2	0.0009
FlyAshPresent	3.6040
SuperplasticizerPresent	7.4810

Although Elastic Net uses a relatively low alpha of 0.2, its coefficient pattern is still similar to Lasso. Compared with OLS, most predictors are shrunk toward zero, while **LogAge**, **FlyAshPresent**, and **SuperplasticizerPresent** retain relatively larger coefficients. No predictor is eliminated, and the coefficient signs remain unchanged. The overall shrinkage is slightly weaker than Lasso, reflecting the combined L1 and L2 penalties in Elastic Net.

1.5 Evaluation

We use RMSE, MAE to evaluate the regression models in this section for the following reasons:

- RMSE: Measures the average prediction error while giving greater weight to large errors, making it useful for detecting models with substantial prediction errors.
- MAE: Measures the average absolute prediction error and provides an intuitive measure of the typical prediction error in the original response scale.

Table 8: Test Set Performance Comparison

Model	RMSE	MAE
OLS	7.1021	5.6944
Ridge	7.2365	5.8099
Lasso	7.0971	5.6942
Elastic Net	7.0974	5.6953

All models are evaluated on the same held-out test set to ensure a fair comparison. Lasso achieves the best performance, with the lowest RMSE of 7.0971 and MAE of 5.6942, followed very closely by Elastic Net and OLS. The differences are very small, indicating that regularization provides only a marginal improvement in generalization for this dataset.

The results also illustrate the bias–variance trade-off. OLS has no regularization, resulting in low bias but potentially higher variance due to multicollinearity. Lasso introduces regularization, slightly increasing bias while reducing coefficient variance, which may explain its marginally better test performance. Ridge produces the highest RMSE and MAE, suggesting that its stronger shrinkage introduces more bias than the reduction in variance can compensate for.

2 Part II - ToothGrowth

2.1 Dataset selection

The built-in **ToothGrowth** dataset was selected because it records a continuous response under every combination of two categorical experimental factors. It contains odontoblast lengths for 60 guinea pigs receiving vitamin C through orange juice (OJ) or ascorbic acid (VC) at daily doses of 0.5, 1, or 2 mg. The six treatment combinations each contain 10 observations, making the data suitable for a balanced two-factor factorial analysis.

2.2 Research question & hypotheses

The response variable is odontoblast length (**len**), in the recorded units of the dataset. The two experimental factors are the vitamin C delivery method (**supp**: orange juice, OJ, or ascorbic acid,

VC) and daily dose (dose: 0.5, 1, or 2 mg/day). Let μ_{ij} denote the population mean odontoblast length for supplement i at dose j . We use a significance level of $\alpha = 0.05$ for all ANOVA tests.

The analysis addresses three related research questions:

1. **Supplement main effect:** Averaged across the three doses, does mean odontoblast length differ between OJ and VC?
 - $H_{0,S} : \mu_{OJ,\cdot} = \mu_{VC,\cdot}$ (no supplement main effect).
 - $H_{A,S} : \mu_{OJ,\cdot} \neq \mu_{VC,\cdot}$.
2. **Dose main effect:** Averaged across the two supplements, does mean odontoblast length differ among the three dose levels?
 - $H_{0,D} : \mu_{\cdot,0.5} = \mu_{\cdot,1} = \mu_{\cdot,2}$ (no dose main effect).
 - $H_{A,D} : \text{at least one marginal dose mean differs.}$
3. **Supplement-by-dose interaction:** Does the effect of supplement on mean odontoblast length depend on dose?
 - $H_{0,SD} : \mu_{OJ,j} - \mu_{VC,j}$ is the same for every dose j (no interaction).
 - $H_{A,SD} : \text{at least one of these supplement differences changes with dose.}$

These hypotheses distinguish the overall effects of the two factors from their interaction. This distinction is important because an overall supplement difference can conceal different supplement effects at individual doses.

2.3 Descriptive statistics

The following code computes both an overall summary and a summary for every supplement-dose cell. Variance and standard deviation use the usual sample definitions. The cell sample sizes also verify that the design is balanced.

Table 9: Overall Descriptive Statistics for Odontoblast Length

N	Mean	Median	Variance	SD	Minimum	Maximum
60	18.813	19.25	58.512	7.649	4.2	33.9

Table 10: Odontoblast Length by Supplement and Dose

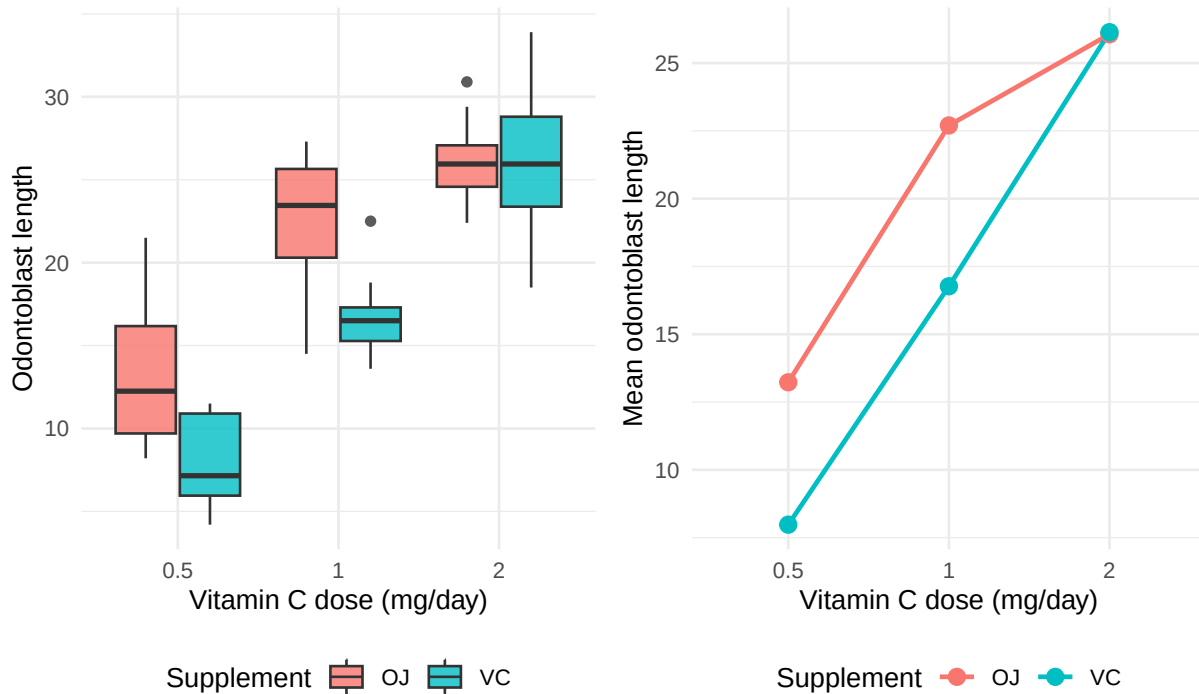
Supplement	Dose	N	Mean	Median	Variance	SD
OJ	0.5	10	13.23	12.25	19.889	4.460
OJ	1	10	22.70	23.45	15.296	3.911
OJ	2	10	26.06	25.95	7.049	2.655
VC	0.5	10	7.98	7.15	7.544	2.747
VC	1	10	16.77	16.50	6.327	2.515
VC	2	10	26.14	25.95	23.018	4.798

There are 60 observations and exactly 10 observations in each of the six cells, so this is a balanced 2×3 factorial design. Overall odontoblast length has mean 18.813 and median 19.250 in the dataset's recorded units. Mean odontoblast length rises with dose for both supplements. For VC,

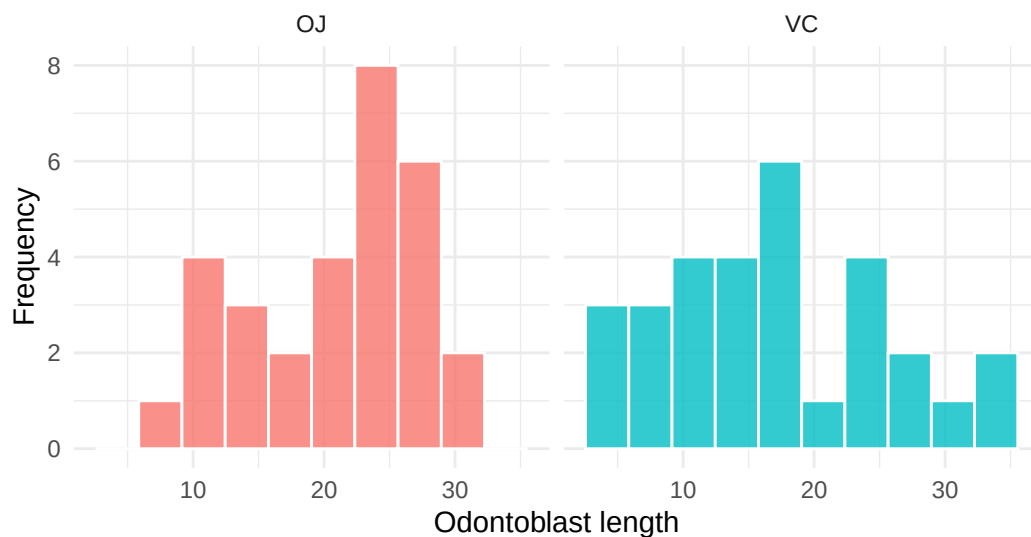
the means at doses 0.5, 1, and 2 mg/day are 7.98, 16.77, and 26.14; for OJ, they are 13.23, 22.70, and 26.06. Thus, OJ has a visibly higher mean at 0.5 and 1 mg/day, while the two supplement means are almost identical at 2 mg/day. This convergence suggests that an interaction may be present and should be tested formally.

The next plots show the distributions, group separation, and mean response profiles from complementary perspectives.

Odontoblast Length by Dose and Supplement Interaction Plot of Cell Means



Distribution of Odontoblast Length by Supplement



The box plots show a strong upward dose pattern for both delivery methods. The interaction plot makes the convergence especially clear: the OJ and VC profiles are separated at 0.5 and 1 mg/day but meet at 2 mg/day. The histograms show the raw response distribution for each supplement; because each histogram pools the dose groups, their spread reflects both within-dose variation and the strong shift in location across doses.

2.4 ANOVA Analysis

2.4.1 Choice of analysis strategy

Among the strategies we have learned, a **two-way factorial ANOVA with interaction** is the appropriate choice. Both predictors are experimentally defined categorical factors, all $2 \times 3 = 6$ treatment combinations are observed, the response is continuous, and the balanced design has 10 observations per cell. The fitted model is

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk},$$

where α_i is the supplement effect, β_j is the dose effect, and $(\alpha\beta)_{ij}$ is their interaction.

A one-way completely randomized design using six combined treatment labels could compare the six means, but it would not directly decompose variation into supplement, dose, and interaction effects. The unequal-sample-size strategy is unnecessary because every cell has the same sample size. Fisher's LSD is a post-hoc comparison method rather than the primary factorial model. ANCOVA would treat dose as a quantitative covariate and impose a particular trend, whereas treating the three assigned doses as a factor allows their means to differ without assuming linearity. Finally, no blocking variable is recorded, so this is a two-way factorial ANOVA rather than a randomized complete block design.

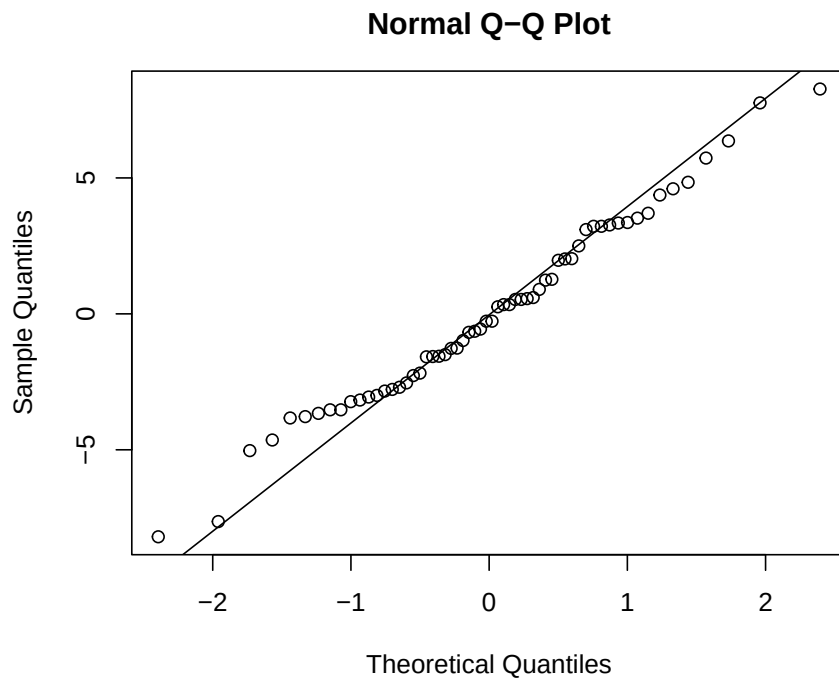
2.4.2 Result

Table 11: ANOVA result

Source	SS	df	MS	F	p_value
supp	205.3500	1	205.3500	15.5720	0.0002
dose	2426.4343	2	1213.2172	92.0000	< 0.0001
supp:dose	108.3190	2	54.1595	4.1070	0.0219
Residuals	712.1060	54	13.1871	NA	NA
Total	3452.2093	59	NA	NA	NA

The ANOVA results show that both supplement type and Vitamin C dose have significant effects on tooth length, with p-values < 0.05 . This indicates that mean tooth length differs between the two supplement types and among the different Vitamin C dose levels. Furthermore, the interaction between supplement type and Vitamin C dose is also significant ($p < 0.05$), indicating that the effect of one factor on tooth length depends on the level of the other factor. In particular, the effect of Vitamin C dose on tooth length differs depending on whether the supplement is administered as vitamin C (VC) or orange juice (OJ).

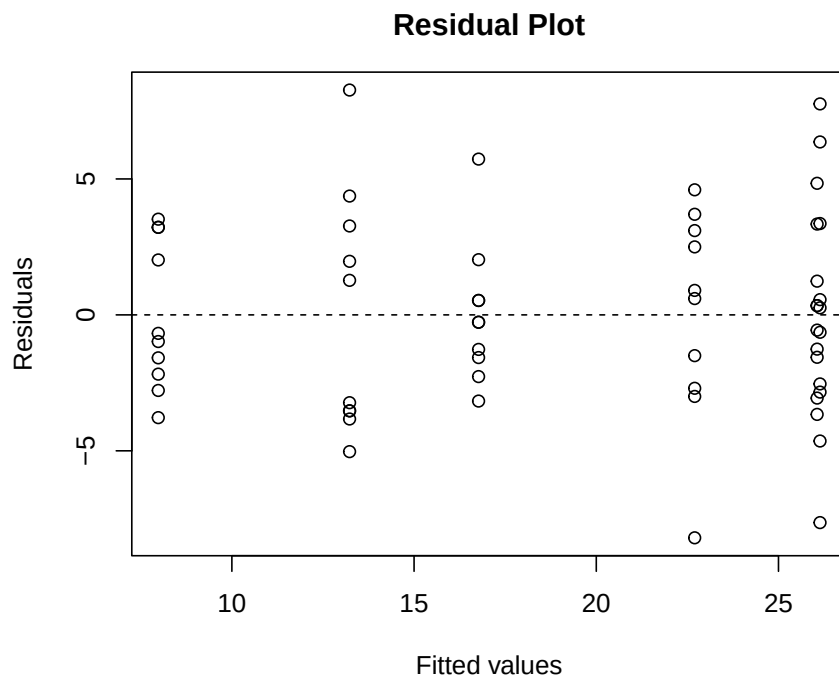
2.4.3 Model adequacy



Shapiro-Wilk Test

W	0.9850
p-value	0.6694

The normal Q-Q plot shows that the points are relatively close to the reference line, suggesting that the residuals are approximately normally distributed. Furthermore, the Shapiro-Wilk test, with the null hypothesis that the residuals are normally distributed, produced a p-value of 0.6694, so we fail to reject the null hypothesis. Therefore, these results suggest that **the normality assumption is satisfied**.

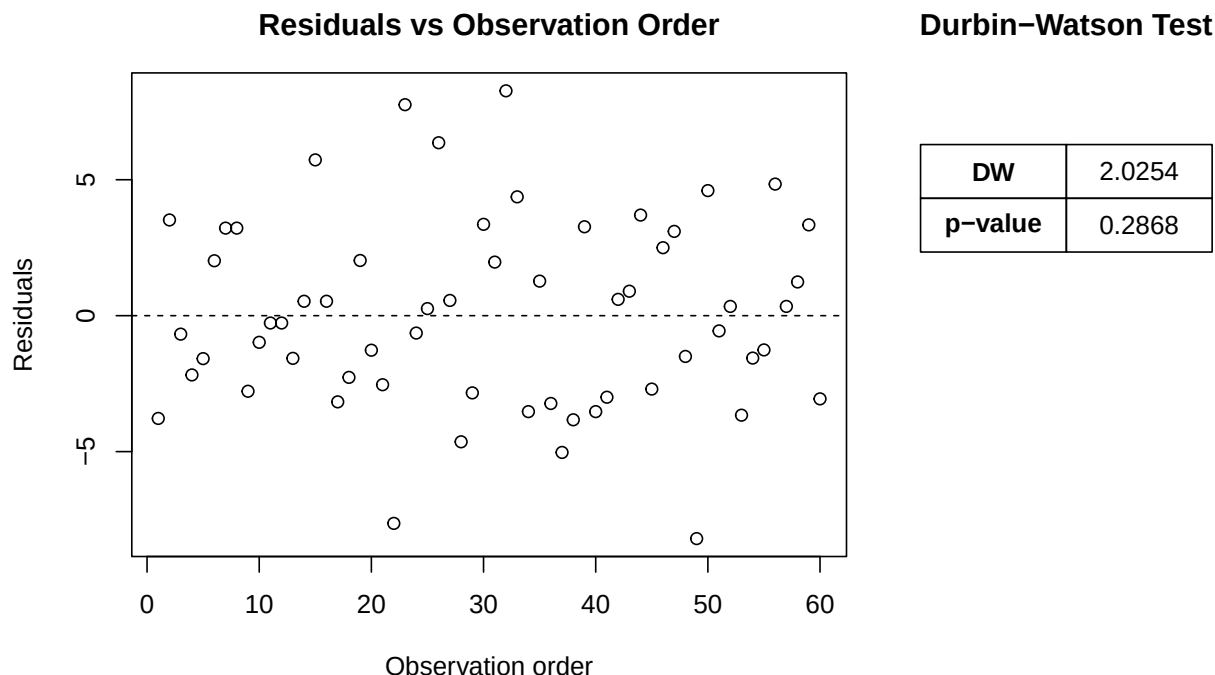


Levene's Test

Df	5
F value	1.7086
p-value	0.1484

By examining the residual plot, we observe that the residuals are distributed around the reference

line with a roughly equal vertical spread across the fitted values, suggesting that the variances are approximately constant. Furthermore, Levene’s test, which has the null hypothesis that all groups have equal variances, produces a p-value of 0.1484. Since this p-value is greater than 0.05, we fail to reject the null hypothesis. Therefore, these results suggest that **the assumption of homogeneity of variance is satisfied**.



When plotting residuals against observation order, the points are randomly distributed around zero with no apparent systematic pattern, suggesting that the residuals are independent. The Durbin–Watson test, which has the null hypothesis that the residuals are independent, gives a p-value of 0.2868. Since $p > 0.05$, we fail to reject the null hypothesis. Therefore, there is no significant evidence of autocorrelation, **supporting the assumption of independence**.

2.5 Post-hoc analysis

The ANOVA results indicate significant effects for both factors and their interaction. However, since supplement type has only two levels, there is no need for a post-hoc pairwise comparison for this factor. Therefore, Tukey’s HSD test will be performed for **dose** and for the **combination of supplement type and dose**.

To interpret the difference between two levels or treatment combinations, we divide their difference by the mean tooth length:

$$\text{diff_mean} = \frac{\text{diff}}{\text{mean of tooth length}}$$

2.5.1 Tukey’s HSD on dose

The results show that all three pairwise comparisons have p-values below 0.05, indicating that the differences between all dose levels are statistically significant. Furthermore, since all differences in means are positive, higher doses of Vitamin C are associated with greater tooth length. Increasing

Table 12: Results of Tukey’s HSD on Dose

	diff	lwr	upr	p adj	diff_mean
1-0.5	9.1300	6.3625	11.8975	< 0.0001	0.4853
2-0.5	15.4950	12.7275	18.2625	< 0.0001	0.8236
2-1	6.3650	3.5975	9.1325	< 0.0001	0.3383

the dose from 0.5 to 2 mg resulted in a great increase of approximately 0.824 times the mean tooth length, indicating a substantial effect of dose.

2.5.2 Tukey’s HSD on interaction between supplement type and dose

Table 13: Results of Tukey’s HSD on Interaction

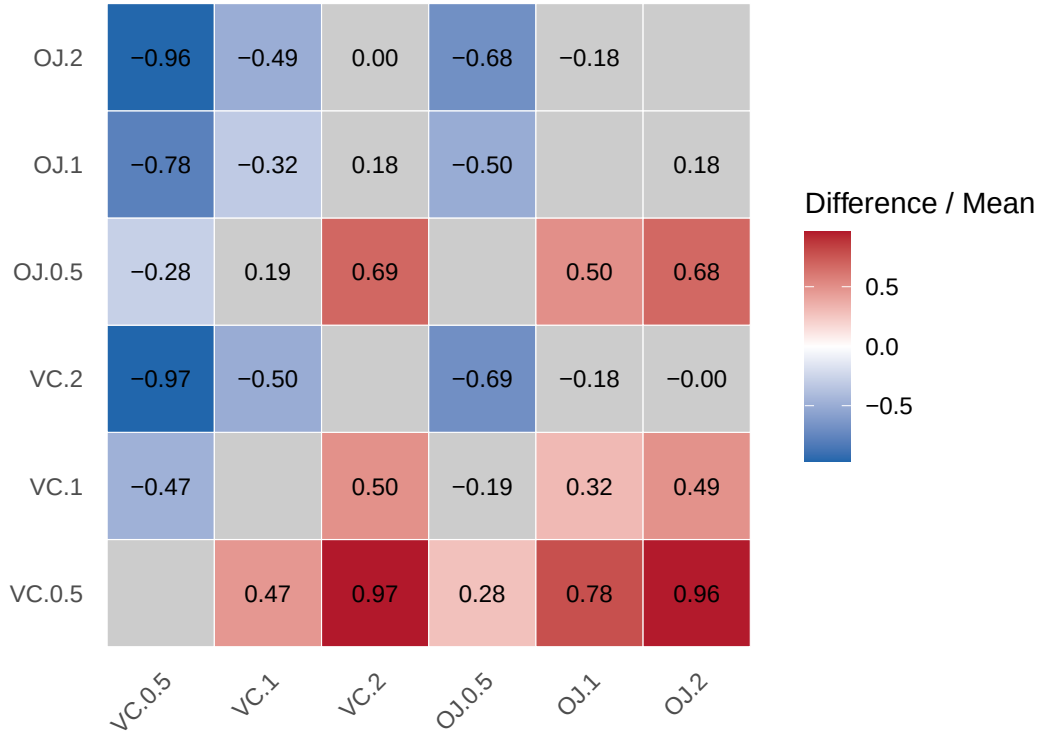
	diff	lwr	upr	p adj	diff_mean
VC.0.5-OJ.0.5	-5.2500	-10.0481	-0.4519	0.0243	-0.2791
OJ.1-OJ.0.5	9.4700	4.6719	14.2681	< 0.0001	0.5034
VC.1-OJ.0.5	3.5400	-1.2581	8.3381	0.2640	0.1882
OJ.2-OJ.0.5	12.8300	8.0319	17.6281	< 0.0001	0.6820
VC.2-OJ.0.5	12.9100	8.1119	17.7081	< 0.0001	0.6862
OJ.1-VC.0.5	14.7200	9.9219	19.5181	< 0.0001	0.7824
VC.1-VC.0.5	8.7900	3.9919	13.5881	< 0.0001	0.4672
OJ.2-VC.0.5	18.0800	13.2819	22.8781	< 0.0001	0.9610
VC.2-VC.0.5	18.1600	13.3619	22.9581	< 0.0001	0.9653
VC.1-OJ.1	-5.9300	-10.7281	-1.1319	0.0074	-0.3152
OJ.2-OJ.1	3.3600	-1.4381	8.1581	0.3187	0.1786
VC.2-OJ.1	3.4400	-1.3581	8.2381	0.2936	0.1828
OJ.2-VC.1	9.2900	4.4919	14.0881	< 0.0001	0.4938
VC.2-VC.1	9.3700	4.5719	14.1681	< 0.0001	0.4981
VC.2-OJ.2	0.0800	-4.7181	4.8781	1.0000	0.0043

Except for the comparisons VC.1 vs OJ.0.5, OJ.2 vs OJ.1, VC.2 vs OJ.1, and VC.2 vs OJ.2, all other pairwise comparisons are statistically significant ($p < 0.05$). The heatmap shows the mean difference in tooth length between the group in each row and the group in each column, with non-significant differences greyed out.

From the heatmap, we can observe that higher doses are generally associated with greater tooth length. Additionally, supplementation with orange juice tends to result in longer tooth length than vitamin C at comparable doses. However, this trend is less consistent when there is a large difference in dose, such as between 0.5 mg and 2 mg, where the effect of dose appears to outweigh the effect of supplement type.

Tukey HSD Relative Mean Differences

Non-significant differences are greyed out



For orange juice supplementation, the variation in tooth length across different doses is less pronounced than for ascorbic acid. In particular, when the dose increases from 0.5 mg to 2 mg, ascorbic acid produces a difference equivalent to 0.97 times the mean tooth length, whereas orange juice produces a smaller difference of 0.68 times the mean tooth length.

2.6 Conclusion and Evaluation

The analysis provides evidence that odontoblast length depends on vitamin C dose, supplement type, and their interaction. Dose is the dominant factor: mean tooth length increases substantially as the dose rises from 0.5 to 2 mg/day. OJ produces greater mean tooth length than VC at the two lower doses, but the two delivery methods have almost identical means at 2 mg/day. The significant interaction therefore shows that the supplement effect must be interpreted at each dose rather than as one constant overall advantage.

The post-hoc results support this interpretation. All marginal dose comparisons are significant, while the treatment-combination comparisons show that several supplement-dose groups are statistically indistinguishable. In practical terms, increasing dose has a stronger and more consistent relationship with tooth length than changing the delivery method, especially when dose differences are large.

The factorial ANOVA is appropriate because the dataset is balanced and contains all six combinations of the two categorical factors. The residual diagnostics and formal tests provide no evidence against the normality, equal-variance, or independence assumptions. Nevertheless, these diagnostics have limited power with 10 observations per cell, and the independence assumption ultimately

depends on how the animals were assigned and measured. The conclusions should therefore be restricted to the dose levels, delivery methods, response measure, and study population represented in `ToothGrowth`; they should not be extrapolated to untested doses or broader populations without additional evidence.

References

Yeh, I-Cheng. 1998. *Concrete Compressive Strength*. UCI Machine Learning Repository.